

The consortium of CoEvolution - **A Comprehensive Trustworthy Framework For Connected Machine Learning And Secure Interconnected AI Solutions** - is proud to announce the official start of this European initiative, funded by the European Commission under the Horizon Europe research and innovation programme.



CoEvolution will create an end-to-end Security, Trust, and Robustness (STR) assessment solution, generating context-aware AI models characterized by their **AI Model Bill of Materials (AIMBOM)**. The goal is a universal hub providing a coherent **STR risk assessment** and security assurance flow, aligning with **MLDevOps** and **EU AI regulatory frameworks**. The paradigm includes novel AI model descriptions, AIMBOM management, security monitoring, and context awareness. CoEvolution introduces a new STR paradigm based on Bills of Materials, offering a unified approach to **describing AI models in supply chains**, ensuring STR compliance with EU directives on trust, fairness, data governance, and **GDPR guidelines**. Open-source trusted datasets and CoEvolution developed AI models enhance the hub's capabilities, aiming for a robust, adaptable risk analysis and security assessment framework aligned with evolving AI cybersecurity threats.

Consortium

CoEvolution is a 3 year Project, coordinated by **Erevnitiko Panepistimiako Institutou Systimaton Epikoionon Kai Ypologiston**, and composed of **15 partners from 8 countries**.



CoEvolution



**A New Horizon
Project for
AI Security,
Trust, and Risk
Assessment in
Machine Learning**

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Pilots

Pilot 1:

Securing the AI Supply Chain of Manufacturing Infrastructure

Goal:

Ensure the security and reliability of AI components in edge computing applications used for industrial automation.

How:

- Develop security mechanisms to verify the integrity of third-party AI components in manufacturing apps.
- Implement trust-building measures to ensure AI models function as expected within complex manufacturing workflows.
- Focus on securing AI models deployed on edge devices.
- Protect edge infrastructure from adversarial attacks.

Pilot 2:

Securing the AI Supply Chain in the Autonomous Vehicles Domain

Goal:

Enhance the resilience and security of CAVs' AI-driven systems ensuring safe autonomous navigation.

How:

- Test cyber-resilience in valet parking scenarios.
- Utilize advanced CAV testing infrastructure, including precise LiDAR, sensors, and high-quality maps.
- Implement cooperative perception for robust environmental modeling and multi-agent path planning.
- Integrate cyber-security techniques to prevent data poisoning and sensor manipulation.



Project's Objectives



Objective 1:

Develop a security and risk assessment process for AI models across the supply chain lifecycle.



Objective 2:

Create a Security, Trust, and Robustness (STR) Defence Framework for AI model security-by-design throughout its lifecycle.



Objective 3:

Propose a new paradigm for AI model security compliant with EU AI guidelines, including an AI Model Bill of Materials as a universal description structure.



Objective 4:

Infuse context-aware mechanisms in AI models to adapt to changing cybersecurity environments.



Objective 5:

Build a secure, trusted AI supply chain ecosystem through the CoEvolution hub, offering secure AI models and datasets.



Objective 6:

Validate and demonstrate the CoEvolution solution in two complex pilots, covering all AI model paradigms.



Objective 7:

Develop a sustainable business plan for the CoEvolution solution post-project.